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Stay Within Your Bounds: Distance-Guided Decoding for Guaranteed Context-Free Grammar Compliance

Lookahead Distance to Acceptance Closes the Infeasibility Gap in Grammar-Constrained LM Decoding

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Grammar-constrained decoding guarantees that a language model's output satisfies a target context-free grammar—enabling reliable generation of JSON, SQL, and logical formulas. But existing methods fail silently: they allow prefixes that no valid completion can extend within the token budget. This paper fixes that gap with a lookahead-guided framework that computes upper-bound distances to the nearest accepting PDA state, uses them to prune infeasible tokens online, and guides beam search toward acceptance. Every output is guaranteed syntactically valid; completion quality improves across all evaluated structure types.

AT A GLANCE

Problem: Grammar masks block individually illegal tokens but permit prefixes from which no budget-feasible completion satisfies the grammar — producing invalid outputs despite the mask.

Why it matters: Hard syntactic guarantees are essential for code generation, structured data extraction, and symbolic planning.

Method: Precompute bounded PDA summaries with upper-bound distances to acceptance; prune tokens that exceed the remaining budget; bias beam search toward acceptance.

Result: 100% syntactic validity on JSON, SQL, and LTL benchmarks; improved completion quality over prior constrained baselines.

Evidence: Experiments across three structure types and multiple grammar sizes; ablations on budget sensitivity and beam width.

The Big Picture

Structured generation tasks require outputs that are syntactically valid instances of a target language: JSON objects, SQL queries, Python expressions, or formulas in linear temporal logic (LTL). Language models produce token sequences autoregressively, and without constraints they generate syntactically invalid outputs at a significant rate.

Grammar-constrained decoding solves this by tracking a pushdown automaton (PDA) derived from the target context-free grammar (CFG). At each step, it masks out tokens that are not consistent with the current PDA state, ensuring only locally valid tokens are sampled. This prevents any single token from violating the grammar.

But local validity is not global validity. A sequence of locally valid tokens can lead the PDA into a configuration from which no complete valid string can be generated within the remaining budget. The decoder emits a prefix that is syntactically valid so far, but cannot be completed.

Why Existing Approaches Fall Short

Two failure modes arise in prior grammar-constrained decoding:

Tokenizer-grammar mismatch: LM subword tokenizers may split strings at positions that do not align with grammar terminal boundaries. A token that spans multiple grammar symbols can leave the PDA in a non-standard mid-rule state that requires specific continuations — which the remaining token vocabulary may not support.

Budget exhaustion: Even with perfect tokenizer alignment, the PDA may enter a configuration requiring many more tokens to reach acceptance than remain in the budget. The decoder continues generating but cannot produce a valid closing sequence.

Both failure modes share the same root: the token mask checks local legality (is this token consistent with the current PDA state?) but not global feasibility (is there any valid completion within budget?).

Core Mechanism

The paper introduces a **distance oracle**: for each PDA configuration $c = (q, \gamma)$ (state q, stack content γ), precompute an upper-bound estimate $\hat{d}(c)$ of the minimum number of additional tokens needed to reach any accepting configuration.

At each decoding step with B tokens remaining, a token a is permitted only if:

1. It is locally consistent with the current PDA state (existing mask)
2. The successor configuration $c' = \delta(c, a)$ satisfies $\hat{d}(c') \leq B$

Condition 2 is the new feasibility filter. It ensures that after token a is emitted, the remaining budget is sufficient to reach acceptance.

Among the feasible tokens, beam search is guided by a distance-adjusted score:

$$s'(a) = \log p_\theta(a \mid x_{<t}) - \lambda \cdot \hat{d}(\delta(c, a))$$

which biases the beam toward tokens that reduce the distance to acceptance, improving completion quality.

Technical Formulation

Let $\mathcal{A} = (Q, \Sigma, \Gamma, \delta, q_0, F)$ be the PDA for the target CFG. Define the **distance to acceptance**:

$$d(c) = \min\Big\{ n \in \mathbb{N} : \exists w \in \Sigma^n,\ c \xrightarrow{w} (q_f, \varepsilon),\ q_f \in F \Big\}$$

with $d(c) = \infty$ if no such w exists.

The paper computes upper bounds $\hat{d}(c) \geq d(c)$ by summarizing PDA reachability with bounded lookahead:

1. For each PDA state q, compute the shortest path in the non-deterministic reachability graph over (q, γ) pairs to any $q_f \in F$, with stack operations truncated at a bounded depth.
2. Propagate distances bottom-up through grammar rule bodies.

The computation is done offline before decoding. Online, only a table lookup per (configuration, token) pair is required.

For a token budget T and a model generating tokens $a_1, \ldots, a_T$:

$$a_t^* = \arg \max_{a \colon \hat{d}(\delta(c_t, a)) \leq T - t} s'(a)$$

guarantees that every emitted prefix can be completed within the remaining budget.

Learning or Inference Procedure

The framework operates purely at inference time:

1. Given a target CFG, compile it to a PDA and compute the distance oracle $\hat{d}(\cdot)$ offline.
2. At each decoding step, compute the set of feasible tokens (locally valid and budget-feasible).
3. Apply beam search with distance-guided scores.
4. Return the highest-scoring complete parse at the end.

No fine-tuning or modification of the base language model is required. The framework integrates with any autoregressive LM.

What the Guarantee Says

The decoder is **sound**: every generated sequence is accepted by the target grammar. Specifically, if the initial budget T satisfies $T \geq \hat{d}(q_0, \perp)$ (the oracle distance from the initial configuration), then at least one feasible token exists at every step and the decoder terminates with a valid output.

The distance oracle guarantee is: for every configuration c encountered during decoding, $\hat{d}(c) \geq d(c)$, so the feasibility filter never incorrectly prunes a token that leads to a valid completion.

Experimental Findings

Evaluated on three structure types with three LLMs (GPT-2-XL, LLaMA-3-8B, Mistral-7B-v0.3):

- **JSON:** 100% syntactic validity for all models and grammar sizes; schema match rate +9.3 pp over greedy and +6.1 pp over token-mask-only baselines.
- **SQL:** 100% parse validity; execution accuracy +7.8 pp over token-mask-only decoding on Spider benchmark.
- **LTL:** 100% formula well-formedness; satisfaction rate of generated plans +11.2 pp over prior constrained baselines.
- **Latency:** Oracle precomputation takes 0.3–1.8 s for practical grammar sizes; online overhead per step is negligible.

Ablations and Interpretation

Budget sensitivity: With tight budgets (T only 10% above $d(q_0, \perp)$), the feasibility filter prunes aggressively and occasionally leaves no feasible token at mid-sequence steps. A budget cushion of 30% eliminates this; the paper recommends budget estimation from grammar depth.

Distance-guidance weight λ: $\lambda = 0$ (pure language model score) recovers the token-mask baseline; $\lambda = 0.5$ achieves the best completion quality. Larger λ over-constrains generation toward acceptance and reduces fluency.

Beam width: Distance-guided beam search with width 4 outperforms greedy distance-guided decoding by +2.1 pp on JSON schema match, with diminishing returns beyond width 8.

Grammar size: The oracle scales well to grammars with up to 500 production rules, covering real SQL dialects. For larger grammars, the bounded-lookahead approximation introduces small oracle errors but validity is preserved.

Reference

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